

Towards Hybrid Modeling of Process Industries with Dynamic Mechanism and Data Augmentation

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Abstract: Modeling and optimization of process industrial plants are critical tasks for continuous improvement in production processes and represent a major direction in the development of industrial intelligence. Due to potential modifications in process industries, such as process upgrades and equipment updates, the accuracy of the predictive model can be compromised, thereby impacting its control and optimization capabilities. In this context, mechanism-based models cannot update parameters to adapt to new characteristics, while data-driven models lack mechanistic constraints. To address these issues, this paper proposes an optimal information content-based data augmentation for model error compensation. By optimizing the information content of transformation functions applied to existing data, the optimal data augmentation functions are selected to construct the error compensation model. Then, the compensation model is integrated with the partially known mechanistic model to obtain the hybrid predictive model that ensures reliable predictions for system control and optimization. The proposed method is validated through simulations on a CSTR system with two system variables and three control variables, and compared with current popular data-driven methods. Experimental results demonstrate that, under certain operational changes, the proposed method maintains higher prediction accuracy and provides better control performance without requiring model updates, validating the effectiveness of the proposed method and its high capability in handling such special scenarios.

Key Words: Optimal Information Content, Model Compensation, Hybrid Model, Process Industries

1 Introduction

Modeling in process industries has always been one of the critical issues considered by both industry and academia [1]. If a predictive model can accurately and realistically reflect the actual changes in a system, our ability to control and optimize industrial systems will achieve a qualitative leap. However, due to the nonlinearity of industrial objects, strong disturbances, and non-negligible sensor errors, accurate modeling is a highly challenging problem. To achieve accurate modeling of system dynamic characteristics, scholars have attempted various approaches. With the rapid development of artificial intelligence methods, data-driven models have become an important research direction [1, 2]. Methods such as artificial neural networks, machine learning, and deep learning have been widely applied to modeling and control. For instance, [3] utilized machine learning methods to optimize and control building energy usage, serving as a typical case. Additionally, mechanism-based methods are also widely used in process industries. These mechanistic models, derived from the initial design, can fully reflect the dynamic characteristics of industrial objects. However, their accuracy can be affected by external disturbances, internal changes, maintenance, and other factors, making long-term operation difficult. Therefore, scholars have shifted their focus to compensating for mechanistic models [4]. For example, [5] designed an adaptive parameter estimation method to compensate for friction disturbances in servo motors, resulting in a more accurate model. [6] used reinforcement learning to update the parameters of an existing turbofan engine mechanistic model, offering a new approach to error

compensation. Furthermore, in the field of model predictive control, [7, 8] proposed schemes to provide disturbance compensation for predictive models, achieving models that better align with actual dynamic characteristics and thus improved control performance. These research outcomes have also spurred the vigorous development of hybrid modeling in process industries.

The aforementioned research has made significant contributions to the modeling of industrial systems, but there are still specific scenarios that have received limited attention from scholars. The problem context considered in this paper originates from field practice, specifically an actual operating unit that, despite having a partially known mechanistic model, cannot be applied to the upgraded unit due to modifications, and thus fails to accurately reflect its dynamic characteristics. However, the mechanistic relationships it contains still capture the primary variable relationships in production. In such cases, re-identifying the mechanistic model alone cannot correctly fit the dynamic characteristics, while common data-driven models, although accurate on the training set, still fail to adhere to mechanistic constraints, exhibit poor generalization, and are prone to generating extreme or non-mechanistic predictions, thereby adversely affecting the control system. Thus, a hybrid modeling framework is more suitable for this problem [9]. Existing hybrid modeling methods, such as [10], compensate for system model errors but do not fully utilize existing data, instead designing complex neural network approaches. Such methods are highly complex and costly to update, especially in model predictive control applications, where there is a potential risk of computational delays. Therefore, designing a low-complexity hybrid modeling method is more appropriate for industrial applications.

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Under this problem context, the author investigated various predictive modeling methods suitable for the issue and ultimately proposed a hybrid modeling approach based on model compensation with data augmentation under optimal information content to address modeling accuracy and improve model generalization in such scenarios. Unlike common system disturbance compensation methods, this paper introduces a framework specifically designed to compensate for the offset errors of partially known mechanistic models. The framework employs a data-augmented linear model to relearn the errors associated with the mechanistic model's offsets. The selection of the augmented dataset is based on maximizing the information content, utilizing the maximized information variation of existing data to train the model. This ensures that the hybrid predictive model under the new framework not only adheres to mechanistic constraints but also accurately adapts to the current production state. Furthermore, the proposed method is applicable to system modeling and model predictive control (MPC) with small datasets. Compared to neural network models that perform well during training, the proposed method demonstrates superior performance when new operating conditions arise. Leveraging the mechanistic constraints inherent in the mechanistic model, the predictive model maintains robust prediction performance, enabling the MPC controller to achieve effective control outcomes.

The structure of this paper is as follows: Chapter 2 provides an explanation of the methods adopted in this study. Chapter 3 introduces the simulation object, simulation design, and simulation results used in this paper. Chapter 4 presents the conclusions and summary of this work.

2 Methodology

2.1 Information-optimal Data Augmentation Method

To enhance the information content of the limited existing data, we analyzed the problem and decided to employ a data transformation method to augment the data, thereby improving its information capacity. Consequently, the author selected 20 transformation functions, as shown in Table1, to transform and expand the dimensionality of the existing data. However, the extensive use of transformation functions introduces significant computational overhead and potential issues of co-linearity. To maximize the data information content while minimizing the number of transformation functions applied, the author designed the following optimization problem. The concept of information entropy is utilized to quantify the data information content, and the optimization problem aims to select the minimal set of transformation functions under the constraint of maximizing information content. The designed optimization problem is as follows:

Define a binary vector $\mathbf{a} \in \{0, 1\}^n$, where $\mathbf{a}_i = 1$ indicates the selection of the i -th transformation function, and $\mathbf{a}_i = 0$ indicates its exclusion. The goal is to maximize the total information content of the transformed dataset while minimizing the number of selected transformation functions.

The optimization problem is represented by Eq1

Table 1: List of Transformation Functions

No.	Trans Function	No.	Trans Function
f_1	$x^2/2!$	f_{11}	$\sinh(x)$
f_2	$x^3/3!$	f_{12}	$\cosh(x)$
f_3	$\sqrt{x}$	f_{13}	$\text{erf}(x)$
f_4	e^x	f_{14}	e^{-x^2}
f_5	$\sin(x)$	f_{15}	$\sin(x^2)$
f_6	$\cos(x)$	f_{16}	$\cos(x^3)$
f_7	$\frac{1}{1+e^{-x}}$	f_{17}	$\tanh(x^2)$
f_8	$\tanh(x)$	f_{18}	$x + x^2 + x^3$
f_9	$ x $	f_{19}	$x \cdot \sin(x)$
f_{10}	$1/x$	f_{20}	$x \cdot \cos(x)$

$$\underset{\mathbf{a}}{\text{minimize}} \quad J(\mathbf{a}) + \|\mathbf{a}\|_1$$

$$J(\mathbf{a}) = - \sum_{i=1}^N \mathbf{a}_i H(f_i(X)) + \sum_{i < j} \mathbf{a}_i \mathbf{a}_j I(f_i(X); f_j(X)) \quad (1)$$

where: $H(f_i)$ is the entropy of the i -th transformation function. $I(f_i; f_j)$ is the mutual information between the i -th and j -th transformation functions. These two information content calculation formulas are shown in Eq2 and 3.

$$H(f_i) = - \sum_{k=1}^K p(f_i = v_k) \log p(f_i = v_k) \quad (2)$$

$$I(f_i; f_j) = \sum_{k=1}^K \sum_{l=1}^L p(f_i = v_k, f_j = w_l) \log \frac{p(f_i = v_k, f_j = w_l)}{p(f_i = v_k)p(f_j = w_l)} \quad (3)$$

$p(f_i = v_k, f_j = w_l)$ is the joint probability distribution of f_i and f_j . $p(f_i = v_k)$ and $p(f_j = w_l)$ are the marginal probability distributions of f_i and f_j , respectively.

The probability density functions are obtained using the kernel density estimation (KDE) method.

We added a sparsity constraint on $\mathbf{a}$ to the optimization problem, ensuring that the selection of transformation functions is sparse.

Through the above optimization problem and solution methods, we can select a combination of transformation functions that maximizes information content while maintaining sparsity. This approach combines information entropy and sparsity, ensuring that data dimensionality is increased while preserving information content and reducing redundancy, resulting in an augmented dataset suitable for modeling in our method.

2.2 Error Compensation Method for Partially Known Mechanistic Models

This method uses existing data and its transformations to construct an augmented dataset, enhancing the characteristics of the limited data. After obtaining the data transformation functions and the transformed dataset, we employ linear regression to predict the model error using the augmented dataset. The reasons for adopting linear regression are as follows: First, data augmentation enhances the information content of the data, enabling even simple models to

perform better. Second, simple models are easy to deploy and computationally inexpensive, making them suitable for industrial scenarios where cost-effective predictive models are required. Third, with the introduction of the mechanistic model, its inherent constraints are naturally satisfied, and only a simple and fast fitting of the errors is needed here.

After completing the data preprocessing, necessary normalization is applied to the data. The prepared data is then used as input to train an Elastic Net regression model to learn and compensate for the errors. Finally, the combined output of the mechanistic model and the error model is used to obtain the final predictive output. The optimization cost function for Elastic Net is shown in Equation 4.

$$\min_{\mathbf{w}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \left(\alpha \|\mathbf{w}\|_1 + \frac{1-\alpha}{2} \|\mathbf{w}\|_2^2 \right) \right\} \quad (4)$$

Where:

- $\mathbf{y}$ is the dependent variable (target variable).
- $\mathbf{X}$ is the independent variable (feature matrix).
- $\mathbf{w}$ is the regression coefficient (weight vector).
- λ is the regularization strength parameter, controlling the overall strength of regularization.
- α is the mixing parameter, balancing the ratio between L1 and L2 regularization:

Here, we denote the obtained data compensation model as $C(x) = \mathbf{X}\mathbf{w}$, and the partially known mechanistic model as $\text{Phy}(x, u)$. Thus, we can derive the hybrid model $H(x, u) = \text{Phy}(x, u) + C(x)$.

Although the existing mechanistic model is not accurate, it provides a mechanistic constraint. After incorporating the error compensation model, the final output becomes accurate and adheres to the mechanistic relationships. Such a model demonstrates better robustness and more accurate predictive performance when facing system variations. We will illustrate the effectiveness of the proposed method through a simulation example.

2.3 Model Predictive Control (MPC)

Model Predictive Control is an advanced control strategy that originated and is widely used in industrial settings. Its optimization control requires a well-performing predictive model as a foundation. The model obtained in this paper serves as the predictive model and needs to be tested within the MPC framework for critical metrics such as control capability and constraint satisfaction. Therefore, the author selected a standard Model Predictive Control problem with Lyapunov constraints for validation, where the Lyapunov constraint is used to ensure the stability of the controller. The basic framework is shown in Eq5:

$$\min_{\mathbf{u}} \sum_{k=0}^{N-1} (\|\mathbf{x}_k - \mathbf{x}_{\text{ref}}\|_{\mathbf{Q}}^2 + \|\mathbf{u}_k\|_{\mathbf{R}}^2) + \|\mathbf{x}_N - \mathbf{x}_{\text{ref}}\|_{\mathbf{P}}^2 \quad (5)$$

Subject to:

$$\mathbf{x}_{k+1} = H(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, 1, \dots, N-1,$$

$$\mathbf{x}_k \in \mathcal{X}, \quad k = 0, 1, \dots, N,$$

$$\mathbf{u}_k \in \mathcal{U}, \quad k = 0, 1, \dots, N-1,$$

$$V(\mathbf{x}_{k+1}) \leq V(\mathbf{x}_k) - \alpha \|\mathbf{x}_k - \mathbf{x}_{\text{ref}}\|^2, \quad k = 0, 1, \dots, N-1,$$

Where:

- $\mathbf{x}_k$ is the state vector at time step k .
- $\mathbf{u}_k$ is the control input vector at time step k .
- $\mathbf{x}_{\text{ref}}$ is the reference state.
- N is the prediction horizon.
- $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{P}$ are positive definite weighting matrices for the state, control input, and terminal state, respectively.
- $H(\mathbf{x}_k, \mathbf{u}_k)$ is the prediction model.
- $\mathcal{X}$ and $\mathcal{U}$ are the feasible sets for the state and control input, respectively.
- $V(\mathbf{x}_k)$ is a Lyapunov function.
- α is a positive constant ensuring stability.

3 Case Demonstration

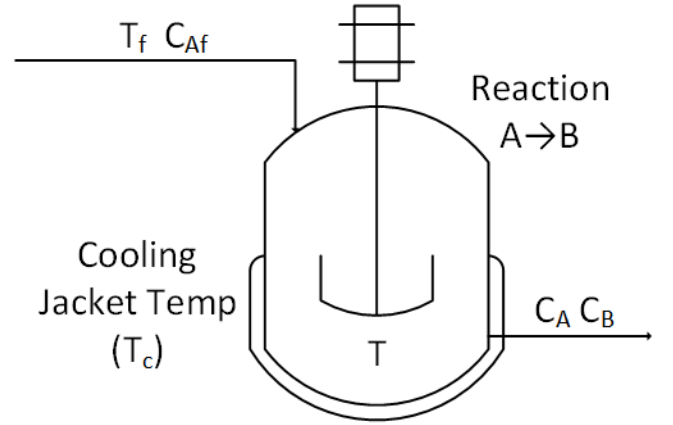


Fig. 1: The CSTR System

This paper uses a CSTR (Continuous Stirred-Tank Reactor) as the simulation object, with the primary focus on two process variables: C_A and T . The three control variables are C_{Af} , T_f , and T_c . A schematic diagram is shown in Figure 1.

The diagram illustrates the CSTR where reactant A is converted to product B. The state variables C_A and T represent the concentration of A and the temperature inside the reactor, respectively.

- C_{Af} is the concentration of A in the feed entering the reactor.
- T_f is the temperature of the feed.
- T_c is the temperature of the cooling jacket used to control the reactor temperature.

The following equations describe the system:

1. **Reaction rate r :**

$$r = K_0 \exp\left(-\frac{E}{RT}\right) C_A \quad (6)$$

2. **Rate of change of concentration $\frac{dC_A}{dt}$:**

$$\frac{dC_A}{dt} = \frac{F}{V}(C_{Af} - C_A) - r \quad (7)$$

3. **Rate of change of temperature $\frac{dT}{dt}$:**

$$\frac{dT}{dt} = \frac{F}{V}(T_f - T) - \left(\frac{\Delta H}{\rho C_p}\right) r - \frac{UA}{\rho C_p V}(T - T_c) \quad (8)$$

These equations describe the changes in reactant concentration and temperature over time in a CSTR. The meanings of the symbols involved in the equations are provided in Table 2. Additionally, some parameter settings are crucial during the simulation, and these parameter details are given in Table 3.

Table 2: Variable Descriptions

Symbol	Description
C_A	Concentration of reactant A
T	Temperature inside the reactor
C_{Af}	Concentration of A in the feed
T_f	Temperature of the feed
T_c	Temperature of the cooling jacket
F	Feed flow rate
V	Reactor volume
R	Gas constant
ΔH	Heat of reaction
E	Activation energy
K_0	Pre-exponential factor
ρC_p	Product of density and heat capacity
UA	Product of heat transfer coefficient and heat transfer area
r	Reaction rate

Table 3: Simulation Parameter Table.

Parameters	Data	Parameters	Data
F	1	R	1.985875
V	1	ΔH	-5960
E	11843	K_0	34930800
ρC_p	500	UA	150
C_{As}	8.5691	T_s	311.2742
C_{Afs}	10	T_{fs}	300
T_{cs}	292		

Using this CSTR as the background, we established operating conditions and corresponding simulation data. By setting different operating points, we obtained 2000 time steps of training data. We then divided this data into an 80% training set and a 20% test set. A set of modeling methods and the proposed method were used as predictive model in the training process, and the model performance was validated on the test set. The model performance is shown in the figure. For comparison, we selected three commonly used models in industrial time-series data modeling: MLP (Multi Layer Perception), SVR (Support Vector Regression), and LSTM (Long Short-Term Memory).

We utilized commonly used error metrics to evaluate the performance of our prediction results. Specifically, we employed Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to assess the prediction errors between the model's predictions and the actual values of the target variable. Additionally, we used the R-squared (R^2) metric to evaluate the fitting capability of the model in capturing the trends of the actual values. The formulas for MAE, RMSE and R^2 are shown below:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (11)$$

In these experimental result figures, Figures 2 and 3 depict the comparison between the predicted curves and reference curves of various methods on the same test set after model training. It can be observed that the performance of each model on the test set is relatively similar, with comparable predictive capabilities. From Table 4, we can see that the proposed method achieves the smallest error and highest accuracy for the prediction of C_A , while from Table 5, MLP performs best for the prediction of T , with the proposed method ranking second. Subsequently, we tested the models on a simulation dataset containing operational condition changes to validate their ability to capture system characteristics under varying conditions. But the simulation data and training data exhibit different characteristics. From Figures 4 and 5, it is evident that SVR's predictive performance deteriorates under new conditions, LSTM's overall predictive capability is weak, and the proposed method and MLP show similar performance for C_A prediction, with the proposed method achieving higher accuracy according to Table 6. Figures 6 and 7 depict the predictions for T , and the results similarly indicate that the proposed method has the strongest predictive capability. To verify the differences in optimization control brought by different models, we also conducted simulations using each model as the predictive model for MPC. Due to the poor prediction performance of LSTM, the optimization problem became infeasible, so no control curves are shown for LSTM. We performed simulations for MLP, SVR, and the proposed method, with the results shown in Figures 8 and 9, representing the control curves for C_A and T , respectively. It can be observed that the proposed model, without requiring online updates, provides control curves with superior tracking performance compared to other methods, better accomplishing control tasks and offering a more reliable and accurate predictive model.

Table 4: Model evaluation metrics on the test set of C_A .

Metrics	MLP	SVR	LSTM	This Method
MAE	0.0012	0.1897	0.0954	0.0002
RMSE	0.0020	0.2294	0.1797	0.0003
R^2	0.9999	0.9398	0.9630	0.9999

Table 5: Model evaluation metrics on the test set of T .

Metrics	MLP	SVR	LSTM	This Method
MAE	0.0025	0.7277	0.3923	0.3884
RMSE	0.0076	0.8312	0.8635	0.3880
R^2	0.9999	0.9481	0.9440	0.9886

4 Conclusion

Through research on process industrial objects containing partial mechanistic models, we found that although the

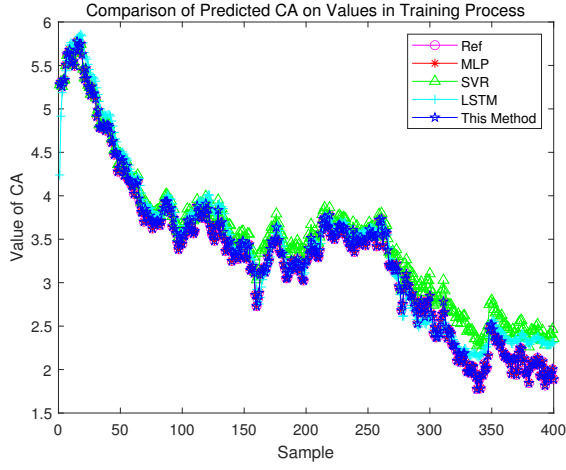


Fig. 2: Comparison of predictions on the test set of C_A .

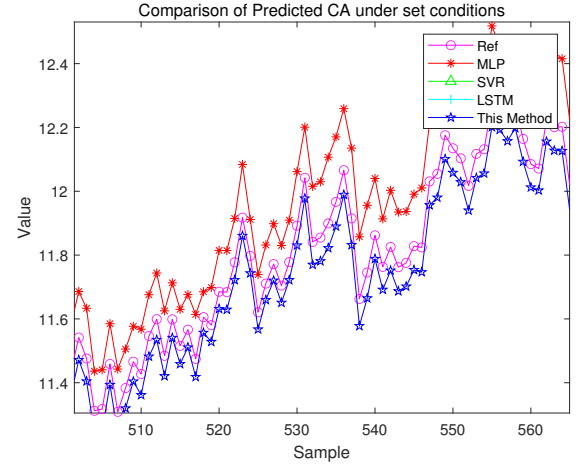


Fig. 5: Detail of Fig4 from time 510-560.

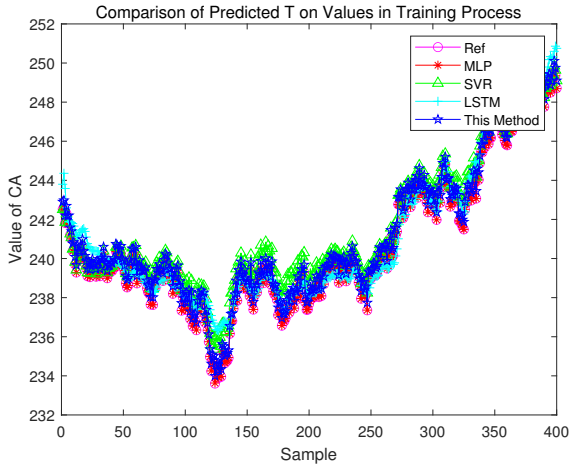


Fig. 3: Comparison of predictions on the test set of T

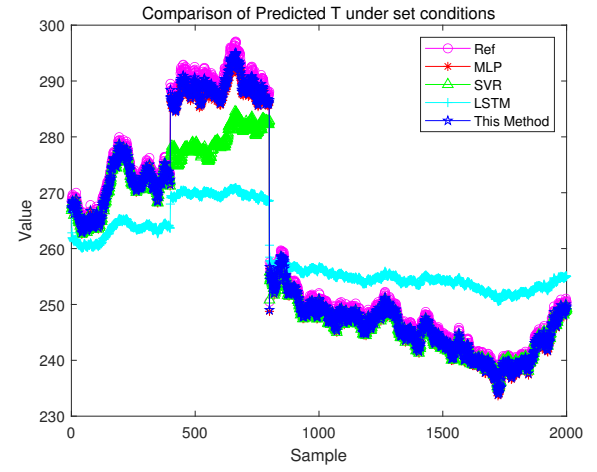


Fig. 6: Comparison of predictions during operating simulation of T .

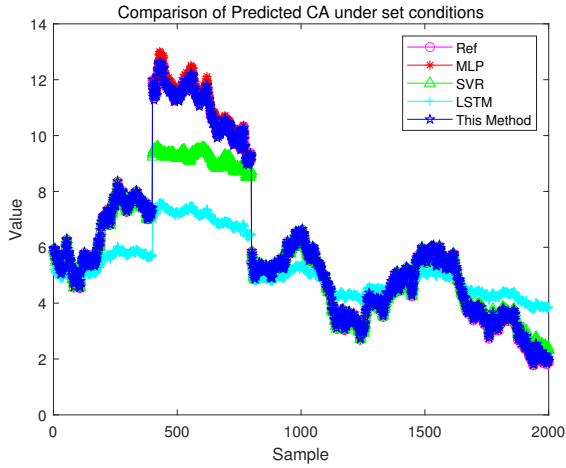


Fig. 4: Comparison of predictions during operating simulation of C_A .

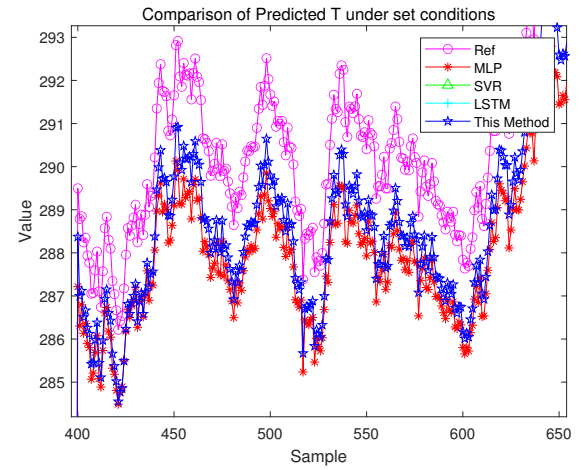


Fig. 7: Detail of Fig6 from time 400-600.

existing mechanistic models cannot provide accurate predictions, the mechanistic information they contain, such as the mechanistic constraints between variables, remains indispensable knowledge. With the goal of efficiently utilizing existing knowledge, the author integrates mechanistic

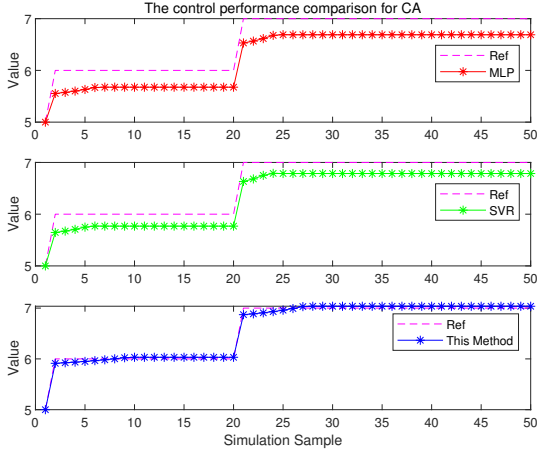
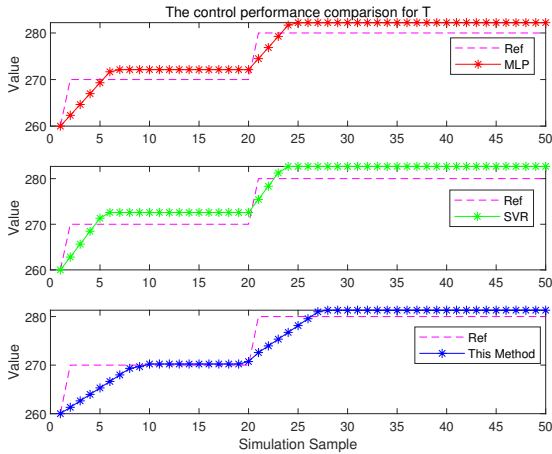
knowledge and data knowledge from process industries from a unified knowledge perspective, constructing the proposed hybrid modeling method. This method combines the data-augmented model error content with partially known mechanistic models, fully leveraging existing object information.

Table 6: Model evaluation metrics of C_A under simulation.

Metrics	MLP	SVR	LSTM	This Method
MAE	0.1586	0.4084	1.4191	0.0335
RMSE	0.1926	0.8680	1.9651	0.0374
R^2	0.9889	0.9034	0.5049	0.9998

Table 7: Model evaluation metrics of T under simulation.

Metrics	MLP	SVR	LSTM	This Method
MAE	1.5202	3.1520	10.7103	1.1795
RMSE	1.5798	5.1698	12.2046	1.2164
R^2	0.9927	0.9223	0.5670	0.9956

Fig. 8: Control curve diagram for C_A .Fig. 9: Control curve diagram for T .

This not only improves the utilization rate of mechanistic models but also enhances the interpretability and acceptability of the hybrid model. Through simulations on a standard CSTR object, this paper thoroughly demonstrates the benefits of the proposed method in predictive modeling and control optimization, providing a solution to the problem under the studied scenario. Of course, there is still value and potential for further research. Combining mechanistic knowledge and data knowledge is a multifaceted issue, involving different expressions, carriers, and applications of mechanistic knowledge, as well as methods and scenarios for integrating data knowledge. The author will continue to explore poten-

tial problems in this context.

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